

Literature resolution: $BV(I)$ embeds strictly into the Riemann–Liouville space $W_{RL,a+}^{s,1}(I)$

Automated Math Discovery Run `fa_banach_001`; Model: GPT5.6
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Status: literature already answered. The open question in arXiv:1603.05033 is answered affirmatively, with strict inclusion, by Theorem 3.4 and Proposition 3.9 of arXiv:2003.09515.

Source question

Let $I = (a, b)$ be a bounded interval and $0 < s < 1$. Bergounioux, Leaci, Nardi, and Tomarelli define

$$W_{RL,a+}^{s,1}(I) = \{u \in L^1(I) : I_{a+}^{1-s}u \in W^{1,1}(I)\}.$$

Their Theorem 4.1 proves $SBV(I) \subset W_{RL,a+}^{s,1}(I)$, but the conclusion states that it was not known whether

$$BV(I) \subsetneq W_{RL,a+}^{s,1}(I) \quad (0 < s < 1).$$

This appears on printed p. 22, Section 5, of the source paper [1, p. 22, Section 5].

Exact later answer

Carbotti and Comi explicitly state that they answer questions posed in the source paper and extend its Theorem 4.1 from SBV to BV .

Answering theorem (Carbotti–Comi, Theorem 3.4 and Proposition 3.9). *For every $u \in BV(I)$ and $0 < s < 1$,*

$$u \in W_{RL,a+}^{s,1}(I), \quad D_{a+}^s u(x) = I_{a+}^{1-s}[Du](x) + \frac{u(a+)}{\Gamma(1-s)(x-a)^s}.$$

The embedding $BV(I) \hookrightarrow W_{RL,a+}^{s,1}(I)$ is continuous. Moreover, $W_{RL,a+}^{s,1}(I)$ strictly contains $BV(I)$.

The embedding and representation are Theorem 3.4, beginning on printed p. 23; the theorem also proves

$$D_{a+}^s u \mathcal{L}^1 \overset{*}{\rightharpoonup} Du + u(a+)\delta_a \quad \text{in } \mathcal{M}([a, b]) \quad (s \uparrow 1).$$

Strictness is Proposition 3.9 on printed pp. 28–29 [2, Theorem 3.4 and Proposition 3.9]. Thus no part of the source’s inclusion question remains open.

Identification and search evidence

The relation is explicit rather than inferred: the answering paper names the source, says it answers questions posed there, and describes Theorem 3.4 as the extension of the source’s SBV theorem to all of BV . Exact-title and phrase searches of the local arXiv source corpus and the official arXiv index found this direct later answer; no originality claim is made here.

References

- [1] M. Bergounioux, A. Leaci, G. Nardi, and F. Tomarelli, *Fractional Sobolev spaces and functions of bounded variation*, Fract. Calc. Appl. Anal. 20 (2017), 936–962; arXiv:1603.05033.
- [2] A. Carbotti and G. E. Comi, *A note on Riemann–Liouville fractional Sobolev spaces*, Commun. Pure Appl. Anal. 20 (2021), 17–54; arXiv:2003.09515.